USAF ACADEMY
DEPARTMENT OF ELECTRICAL AND COMPUTER ENGINEERING
LESSON 22–23 LAB - Tapering, Squint, and Quantization
Lab Sheet
Fall 2026

Name: _____ **Section:** _____ **Date:** _____

Part 1 – Learning Objective 3.7: I can apply amplitude tapering (uniform, cosine, Chebyshev, Taylor) to control sidelobe level and predict the pattern trade-off.

This sheet is the turn-in document for both parts of the L22–L23 Lab. Predict each row before you sweep it. In Part 1, keep the peak drop and the taper efficiency in separate columns throughout. The lab is individual: one sheet per cadet, from your own kit.

1. **The taper table: the element gains.** Read the eight slider values off Element Gains for each preset and record them as percentages.

Preset	Rx1	Rx2	Rx3	Rx4	Rx5	Rx6	Rx7	Rx8
Uniform								
Hann								
Blackman								
Chebyshev								

The taper table, continued: what the sweep reads. Predict each row before you press Start. The peak drop is measured against the frozen uniform trace, and $\eta_t = (\sum a_n)^2/(N \sum a_n^2)$ is computed rather than measured.

Preset	HPBW pred	HPBW meas	Drop pred	Drop meas	First sidelobe	η_t	η_t (dB)
Uniform							
Hann							
Blackman							
Chebyshev							

2. **Your custom taper.** Record the eight gain values you settled on and the measured result. The specification was a half-power beamwidth no wider than 17° with the first sidelobe below -20 dBc.

Taper	Rx1	Rx2	Rx3	Rx4	Rx5	Rx6	Rx7	Rx8
Custom								

Measured HPBW = Measured first sidelobe =

In one sentence, how did you arrive at these values?

3. Written answers. A short paragraph each.

(a) The tapered sidelobes disappear from the plot, but the beamwidth change is easy to see and easy to measure. Explain why the measurement gives a good number for one and no number at all for the other.

(b) Explain why the peak drop you measured is not the directivity the array lost, and what each number would be used for.

Part 2 – Learning Objective 3.8: I can identify beam squint and quantization effects in a phased array and describe their impact on system performance.

Lesson 23 derived the three departures no taper can fix; here you measure them. Predict each row before you sweep it, and use the Signal Freq the GUI shows for λ throughout (about 10.5 GHz, so $\lambda \approx 28.5$ mm).

4. **Grating lobes.** Source at boresight, Enforce Symmetric Taper off. Lab preset 4 leaves Rx1, Rx4, and Rx7 on, so the active spacing is $d_{\text{eff}} = 3d = 42$ mm. Then set Rx4 and Rx7 to 0 and Rx5 to 100 for $d_{\text{eff}} = 4d = 56$ mm. A full-height lobe appears at every angle where $\sin\theta = m\lambda/d_{\text{eff}}$ has a real solution. Predict the angles before each sweep. The peak drop is read against the frozen uniform trace, and its calculated value follows from the number of elements still on.

Elements on	d_{eff}	Lobes calc	Lobes meas	Height vs 0° lobe	Drop calc	Drop meas
All eight	14 mm					
Rx1, Rx4, Rx7	42 mm					
Rx1, Rx5	56 mm					

In one sentence: every grating lobe stands as tall as the main lobe, and no sidelobe ever does. What is the difference between the two?

5. **Beam squint.** Move the HB100 to about 45° on the protractor arc and press the Uniform preset. Lab preset 5 loads Signal BW = 500 MHz: the GUI computes the steering phases for a frequency 500 MHz below the Signal Freq and measures at the Signal Freq, which is what a signal at the edge of a 500 MHz band sees. Set Signal BW to 10 MHz, sweep, read Est. Angle, and press Freeze. Then set it back to 500 MHz and sweep again. The calculated shift comes from the squint relation in Lesson 23: a beam commanded to θ_0 with its phases set at f_0 points at $\sin^{-1}[(f_0/f)\sin\theta_0]$ when observed at f.

Signal BW	Est. Angle meas	Shift calc	Shift meas
10 MHz			
500 MHz			

The source did not move between the two sweeps. In one sentence, what did?

6. **Phase quantization.** Source back at boresight. Lab preset 6 loads the Blackman taper, so every true sidelobe is below the floor and anything that appears is the phase shifter's doing. Turn **Use Bits** off, so the sweep keeps stepping the commanded angle by 2.8125° while the Phase Shift Bits slider coarsens the element phases. Sweep at 7, 4, 3, and 2 bits. The Blackman main lobe is about 24° wide and its skirt reaches the floor near $\pm 35^\circ$, so record the highest lobe *beyond* $\pm 35^\circ$ relative to the peak, and beside it the Lesson 23 rule of thumb for the quantization sidelobe level, $-6B$ dB.

Bits B	**LSB calc**	**Highest lobe beyond $\pm 35^\circ$, meas**	**$-6B$ dB**
7			
4			
3			
2			

The rule of thumb puts the 7-bit lobes near -42 dB. In one sentence, why does that row have no number, and what is the deepest level this sweep could report for this taper?

Before you leave: Use Bits on, Phase Shift Bits 7, Signal BW 10 MHz, and the Uniform preset.

Documentation: _____